

Air Fryer Demand & Pricing Analysis

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Brand-year demand estimation from Amazon purchase data, 2019–2023

Main takeaway

Ninja and Instant Pot look like the strongest economics in this market: high share, healthy markups, and strong demand positioning.

10

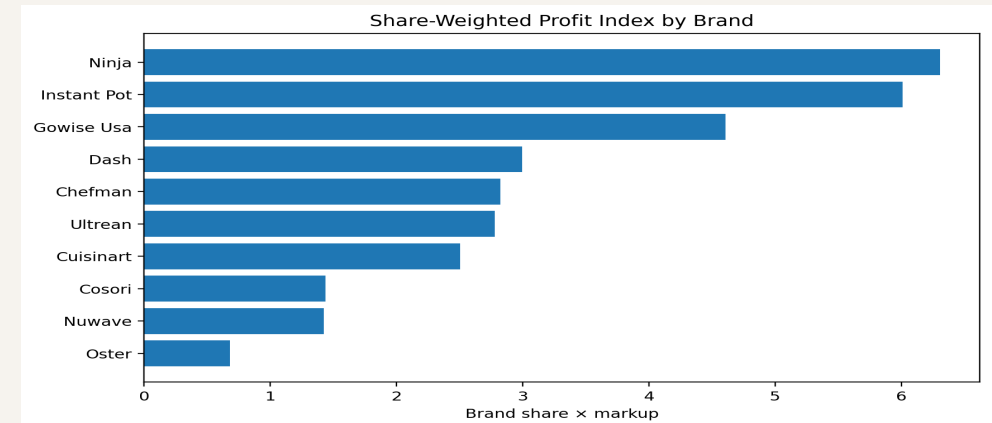
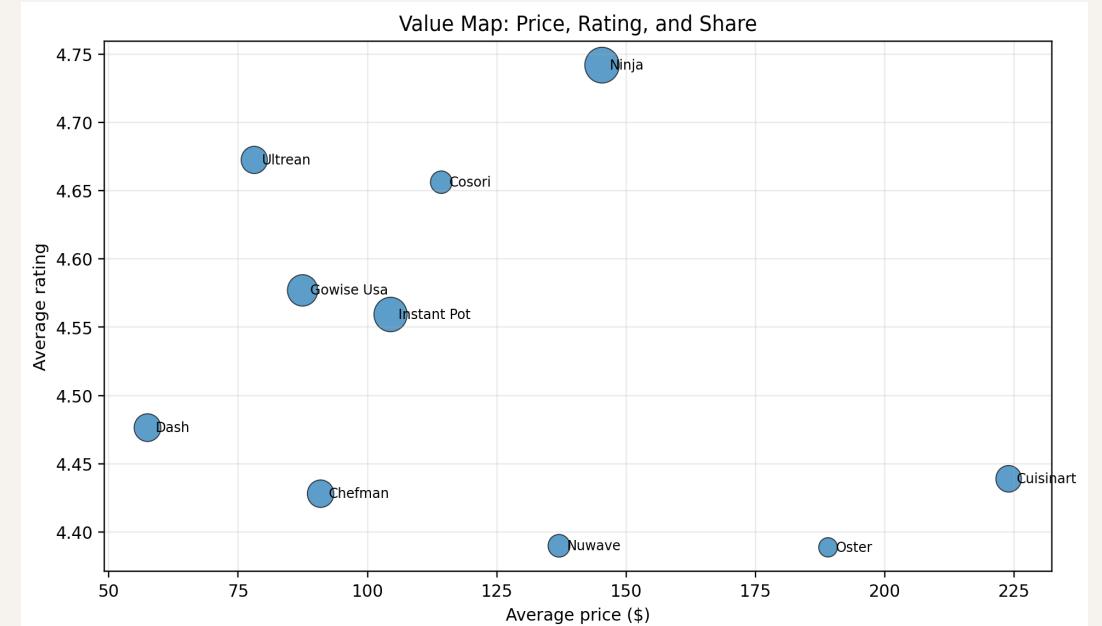
brands

50

brand-year rows

0.76

model R^2



1) Data Analyst View: the market is measurable and share-normalized

Each row is a brand-year observation for the top included air-fryer brands.

2019–2023

analysis window

1.00

share sums within each year

10

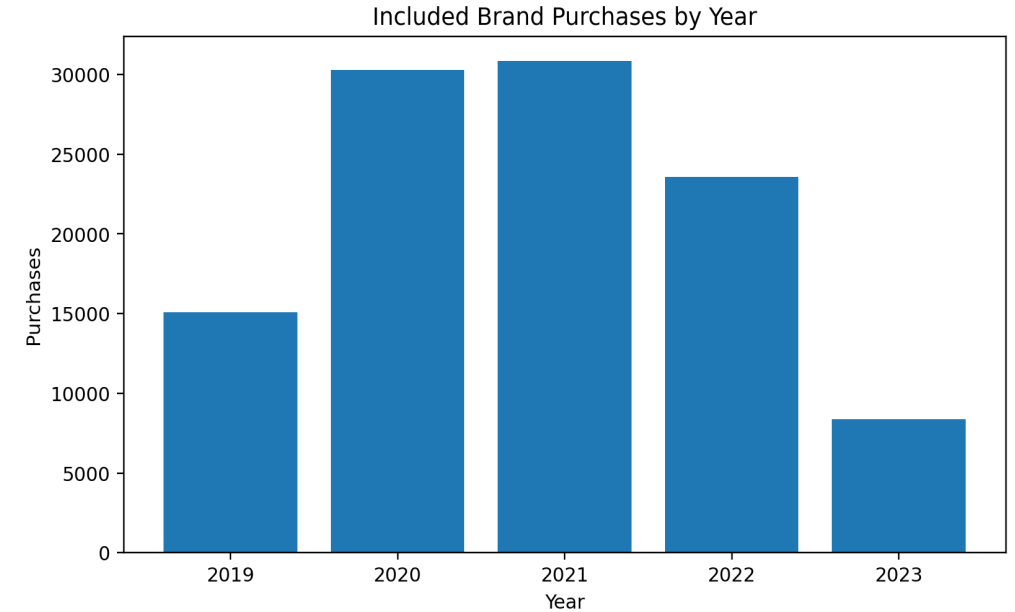
brands tracked

What this means

The dataset is balanced: 10 brands × 5 years = 50 observations.

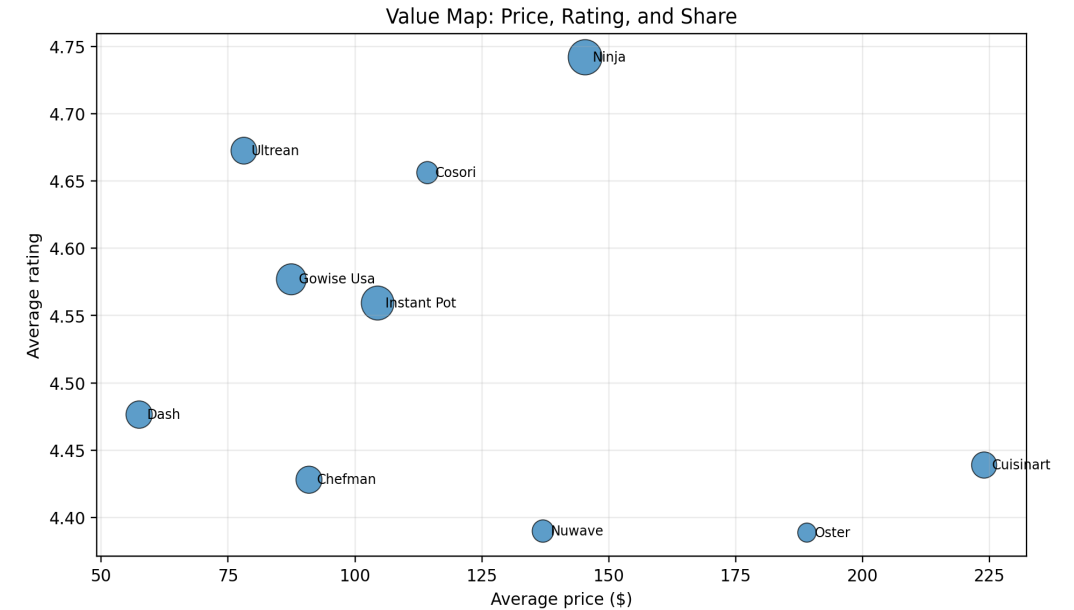
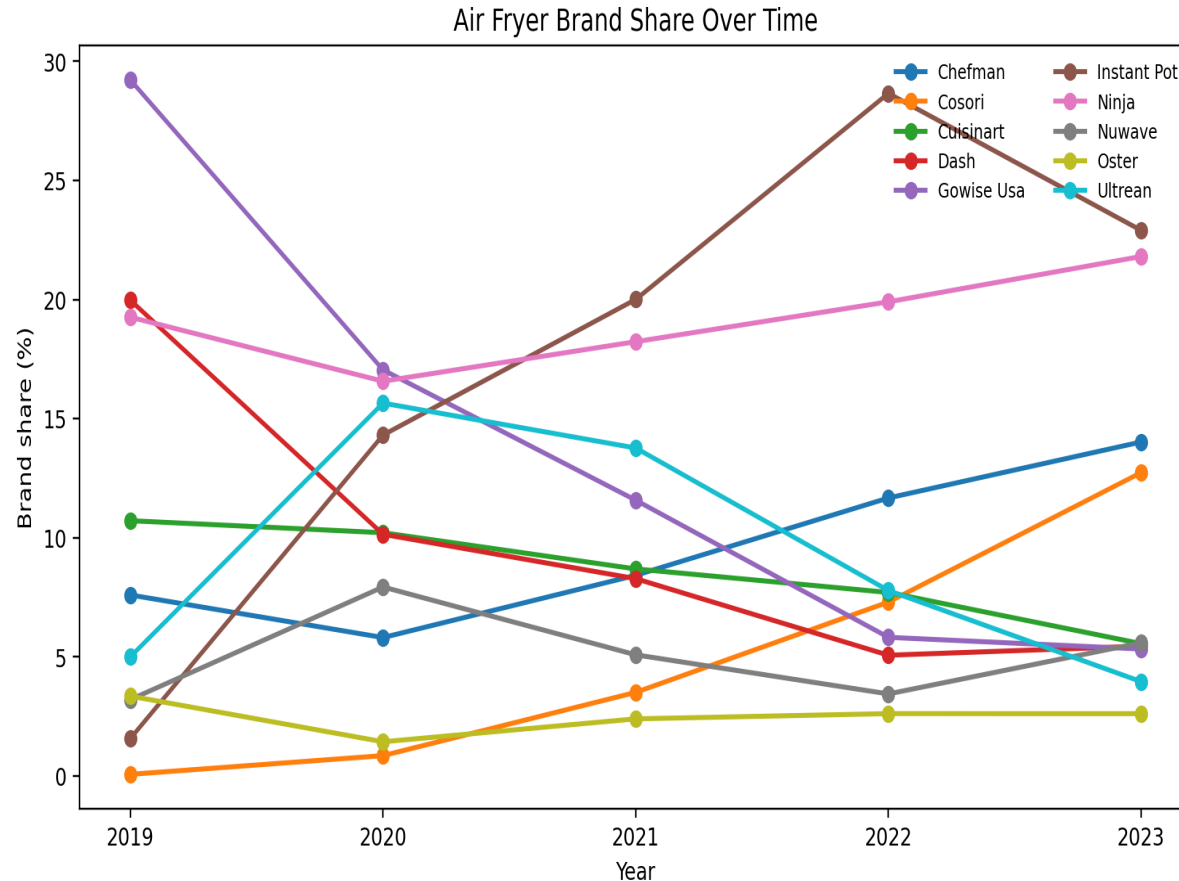
Market share is normalized inside the included air-fryer brands, so the analysis is about competition within this brand set.

Purchases peaked around 2020–2021, then fell in 2023 in this cleaned sample.



2) EDA: Ninja and Instant Pot carry the strongest share positions

Share leadership is concentrated, while price tiers vary widely across brands.



Business read

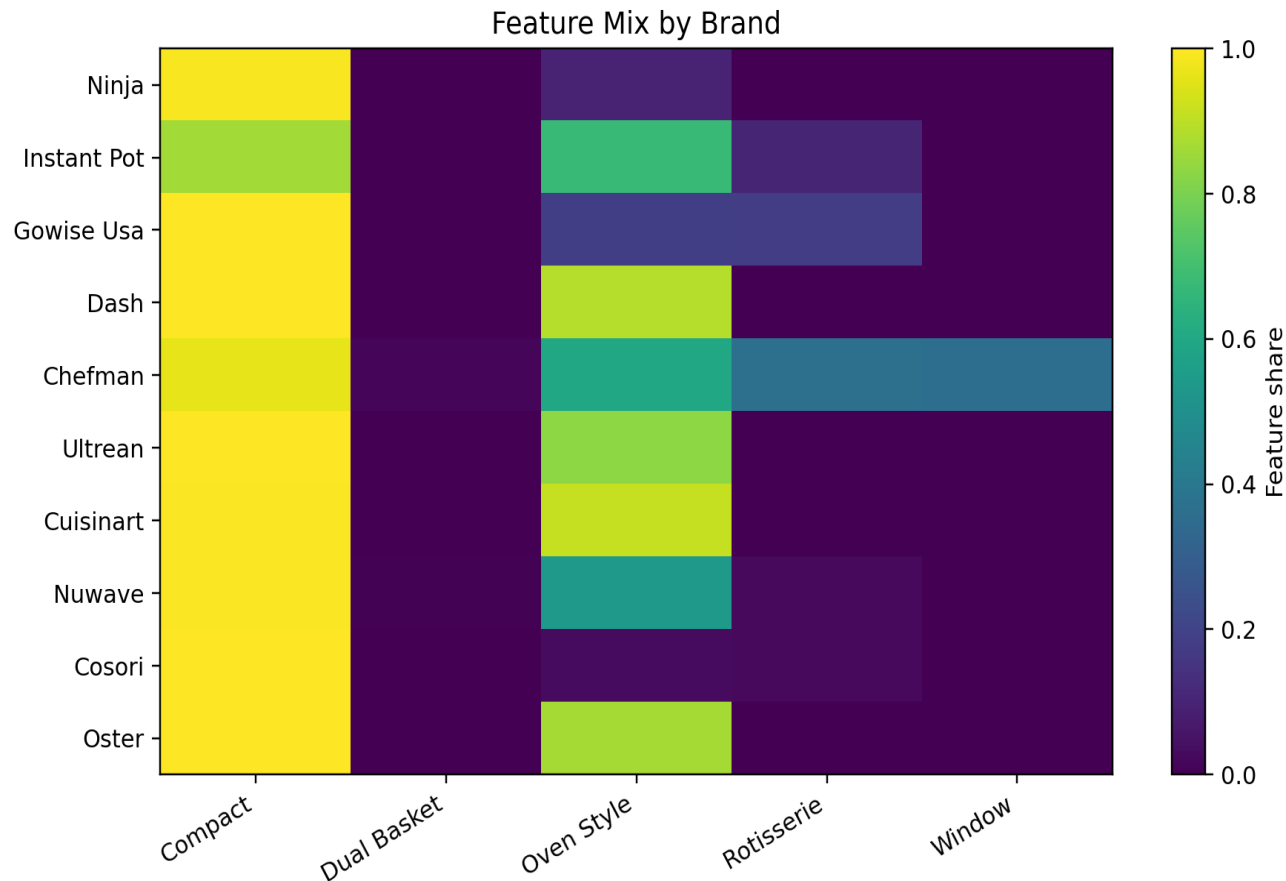
Ninja has the best combination of share and rating.

Cuisinart and Oster are premium-priced, but smaller in share.

Dash is lower-priced, suggesting a value/entry-tier position.

3) Product positioning: feature mix differs by brand

Compact designs are nearly universal; specialty features are concentrated.



Feature interpretation

Compact is common across almost every brand, so it is not enough by itself to differentiate.

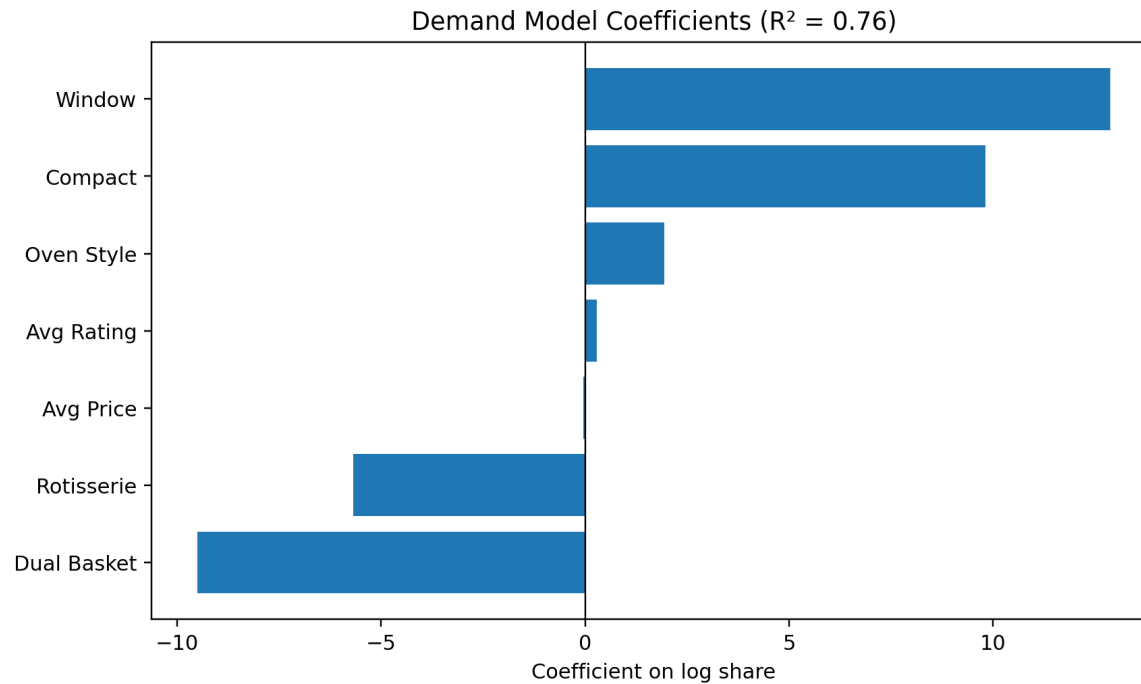
Oven-style products are common for Dash, Cuisinart, Oster, and Ultrean.

Chefman stands out on rotisserie and window-style features, indicating more visible specialty positioning.

Dual-basket is rare in this sample, so its coefficient should be interpreted carefully.

4) Demand model: price moves demand in the expected direction

Regression uses price, rating, feature shares, brand dummies, and year dummies.



-0.0377

price coefficient

+0.2875

rating coefficient

0.76

3

R^2

Interpretation

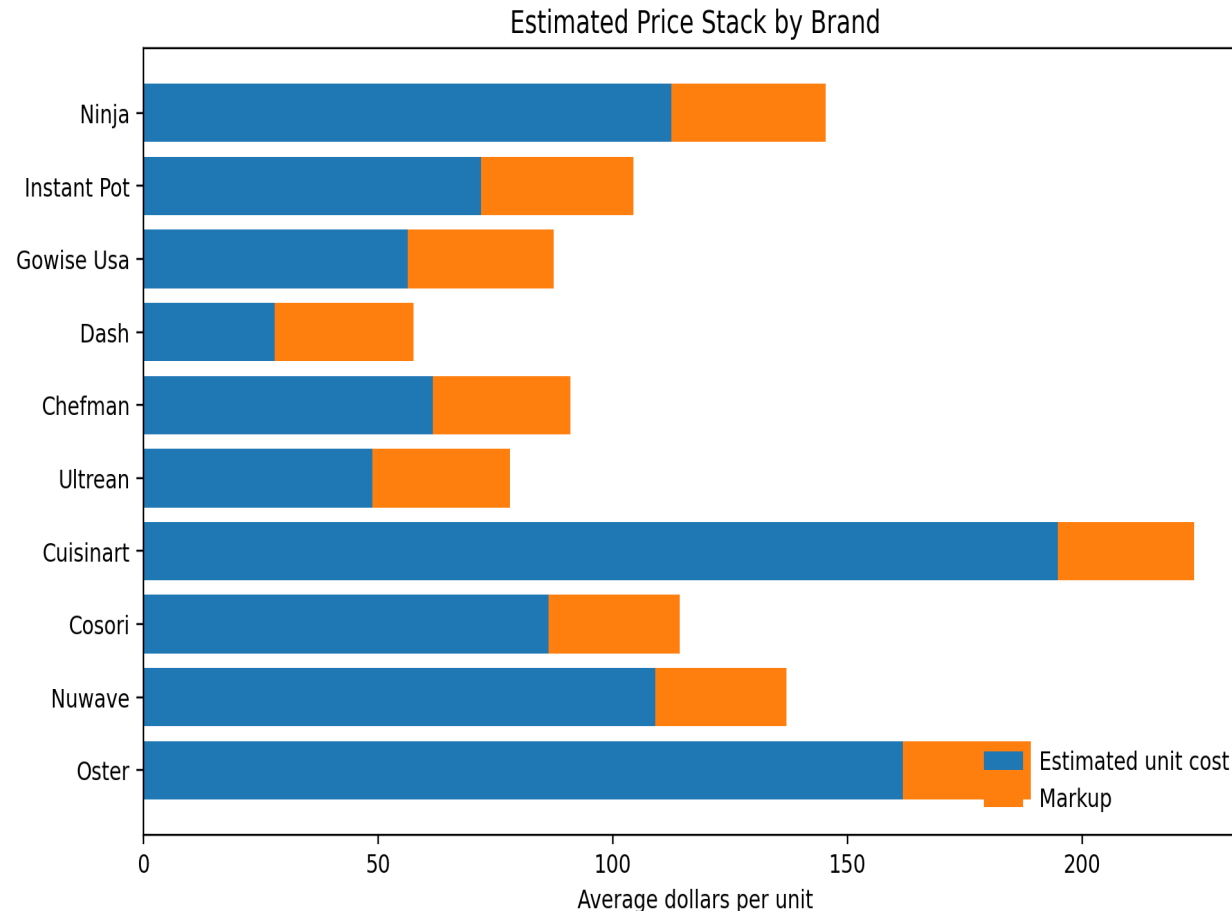
The negative price coefficient is important because it makes the pricing-analysis step economically sensible.

Higher ratings are associated with stronger brand share, holding other variables constant.

Feature coefficients are useful directionally, but should not be overread because the sample is small and features are tied to brand strategy.

5) Pricing economics: no negative costs; markups are fairly consistent

Demand estimates imply unit cost, markup, and a share-weighted profit index.



What stands out

Estimated unit costs are all positive, so the model does not produce an obvious economic red flag.

Cuisinart and Oster have the highest implied costs because their price levels are highest.

Markup levels cluster around roughly \$27–\$33, so profit differences mainly come from share differences.

\$32.85

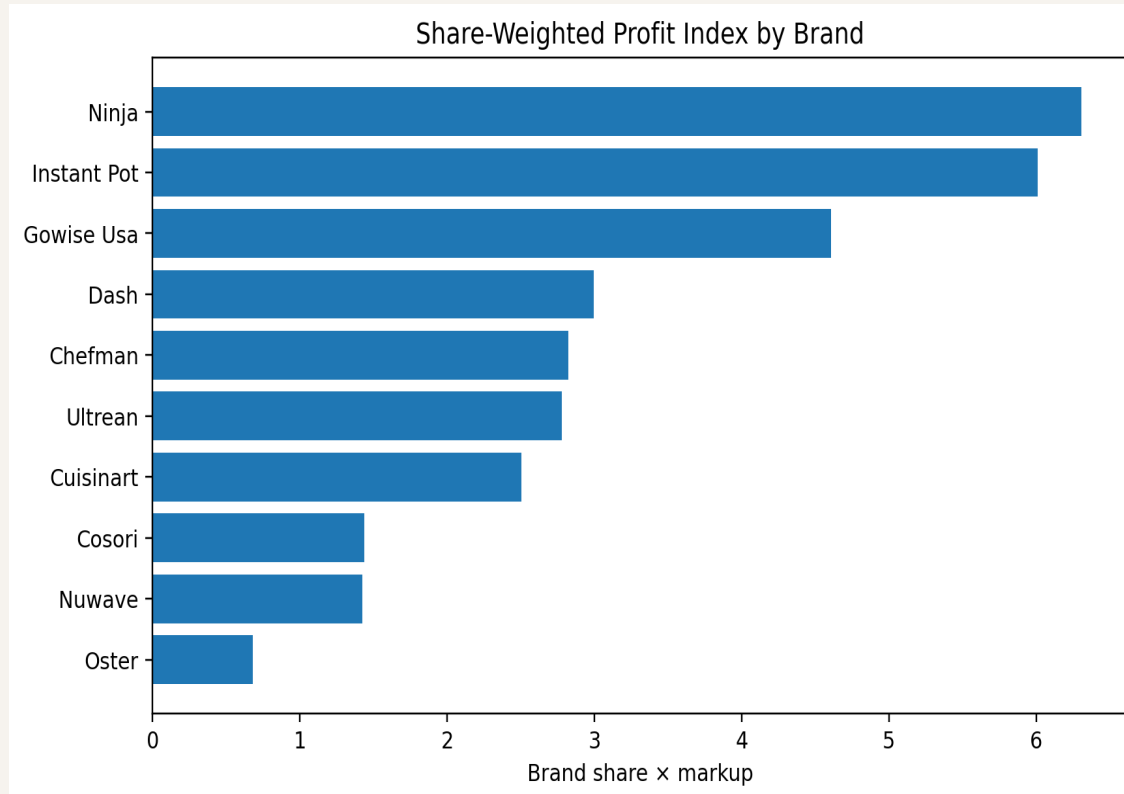
avg. Ninja markup

\$32.56

avg. Instant Pot markup

6) Recommendation: protect the leaders, differentiate the rest

Pricing strategy should account for both demand strength and brand positioning.



Priority actions

- 1 Protect Ninja and Instant Pot price architecture; they are the clearest profit engines.
- 2 Use features and ratings to justify price rather than relying only on discounting.
- 3 For premium brands, focus on niche positioning; high prices do not automatically translate into high profit contribution.

Final takeaway

The best market position is not simply the highest price or the lowest price it is the combination of brand share, consumer value, and sustainable markup.